# Chapter 12: Linear Regression and Correlation

## Introduction

In this chapter, we will be studying the relationship between **two quantitative variables**.

Some real-world areas that use linear regression to understand the relationship of two or more variables:

- Business - the relationship between advertising spending and revenue.
- Medical research - the relationship between drug dosage and blood pressure of patients.
- Agricultural Science – to measure the effect of fertilizer and water on crop yields.
- Data Science – the effect that different training regimens for professional sports teams have on player performance.

We will progress from graphical displays to numerical summaries and then proceed to mathematical models and statistical inference.

We will use three tools to describe, picture, and quantify the relationship between **two** quantitative variables:

- **Scatter Plot** - a two-dimensional graph of data values
- **Correlation** - a statistic that measures the *strength and direction* of a linear relationship between two quantitative variables.
- **Regression equation** - an equation that describes the average relationship between a quantitative response variable and an explanatory variable.
    - $\hat{y} = b_1 x + b_0$ where $b_1$ is the slope and $b_0$ is the y-intercept.

## 12.1: Linear Equations (Algebra)

In the equation $y = ax + b$

- $a$ represents the **slope** and
- $b$ represents the $y$ **- intercept**.

The **slope** of a line is defined by the change in $y$ divided by the change in $x$.

The $y$ **- intercept** is the value of $y$ when $x = 0$.

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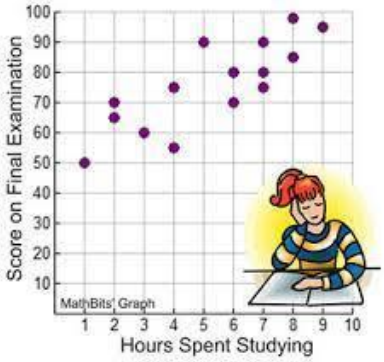
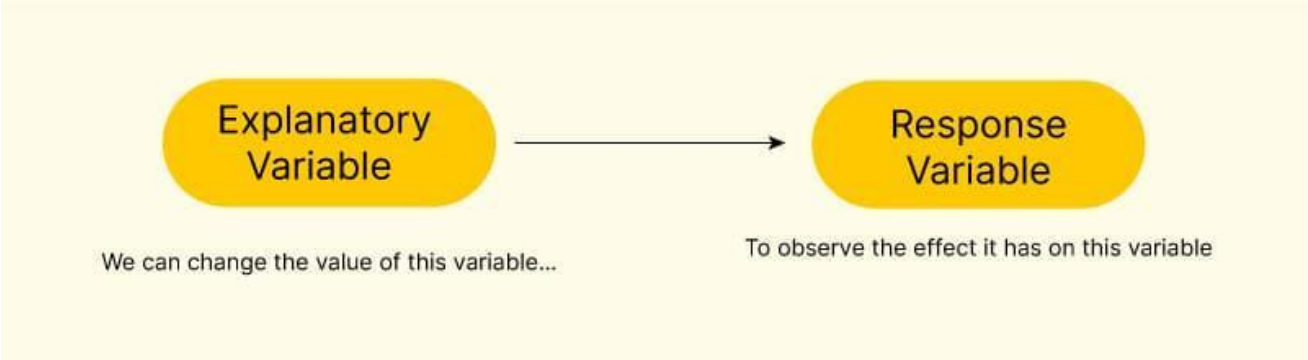
# 12.2: Scatter Plots

**Scatter Plot** - a two-dimensional graph that shows the relationship between two quantitative variables measured on the same individual. Each individual in the data set is represented by a point in the scatter diagram.




**Explanatory variable** or **predictor variable** - plotted on the horizontal (*x*) axis, used as input, and can help explain another variable. It is also known as the **independent variable**.

**Response variable** - plotted on the vertical (*y*) axis whose value can be explained by the explanatory variable. It is also known as the **dependent variable**.




Two variables that are <u>linearly related</u> are said to be **positively associated** if, whenever the value of one variable increases, the value of the other variable also tends to increase. (as *x* increases, *y* tends to increase)

Two variables that are <u>linearly related</u> are said to be **negatively associated** if, whenever the value of one variable increases, the value of the other variable tends to decrease. (as *x* increases, *y* tends to decrease)

Two variables have a **linear relationship** when the pattern of their relationship resembles a straight line.

If a scatterplot shows a curved line instead of a straight line, the relationship is called **nonlinear** or **curvilinear**.

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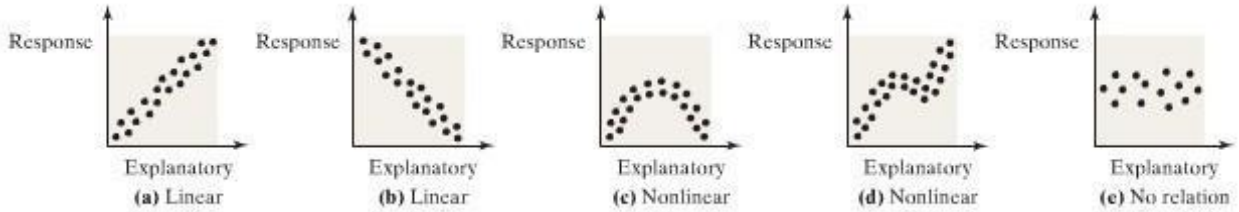
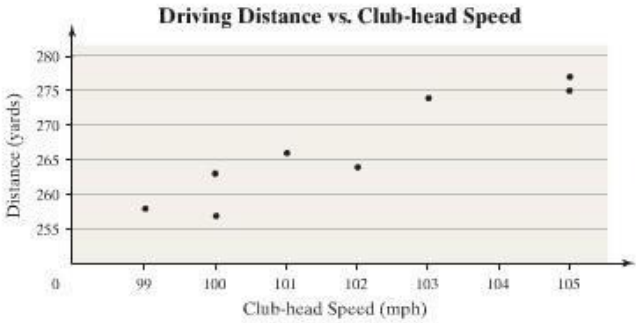
A few examples of scatter plots:




Example 1. A golf pro wanted to investigate the relation between the club-head speed of a golf club (measured in miles per hour) and the distance (in yards) that the ball will travel. He realized that there are other variables besides club-head speed that determine the distance a ball will travel (such as club type, ball type, golfer, and weather conditions). To eliminate the variability due to these variables (known as lurking variables), the pro used a single model of club and ball. One golfer was chosen to swing the club on a clear, 70-degree day with no wind. The pro recorded the club-head speed, measured the distance that the ball traveled, and collected the data below. Draw a scatter diagram of the data.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Club-Head Speed (mph) | 100 | 102 | 103 | 101 | 105 | 100 | 99 | 105 |
| Distance (yards) | 257 | 264 | 274 | 266 | 277 | 263 | 258 | 275 |




What is the explanatory variable and its unit of measure?

What is the response variable and its unit of measure?

Do you think the two variables are linearly related? Is it in a positive or negative direction?

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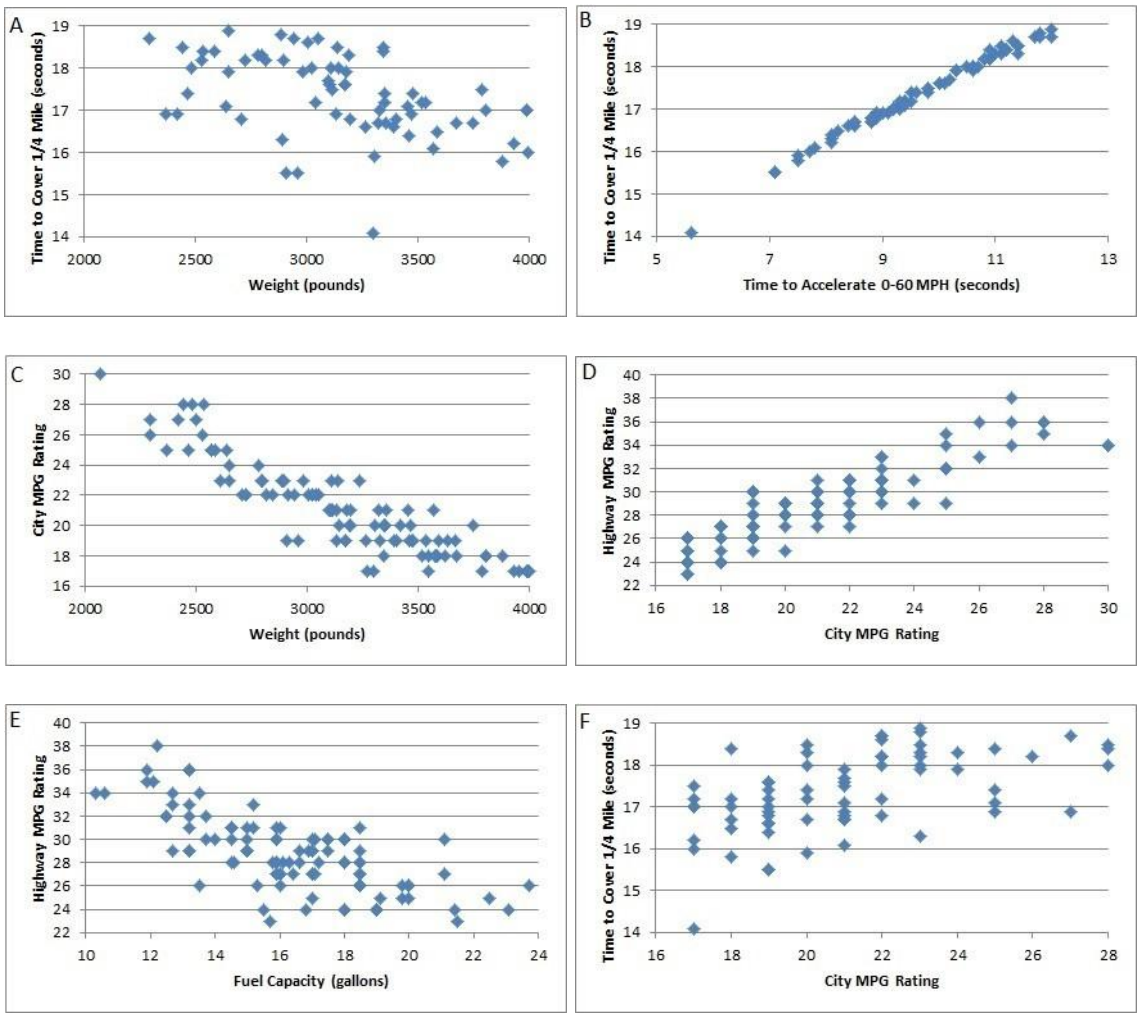
Example 2. Let us evaluate the strength (weak, moderate, strong) and direction (positive or negative) of the linear association between the two variables in each graph.




## Correlation Coefficient "r"

The statistical **correlation between** two quantitative variables is a number that *indicates the strength and the direction of a straight-line relationship*.

- The *strength* of the relationship is determined by the *closeness of the points to a straight line*.
- The *direction* is determined by whether one variable <u>generally</u> increases or <u>generally</u> decreases when the other variable increases.

The **linear correlation coefficient "r"** is a measure of the <u>strength</u> and <u>direction</u> of the linear relation between two quantitative variables.

- We use the Greek letter $\rho$ (rho) to represent the population correlation coefficient.
- We use "r" to represent the sample correlation coefficient.

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**What the VALUE of "r" (sample correlation coefficient) tells us:**

- The value of "r" is always between −1 and +1: $-1 \leq r \leq 1$

- The *magnitude* of the correlation "*r*" indicates the strength of the linear relationship between *x* and *y*, which is the overall closeness of the points to a straight line. The *sign* of the correlation does not matter when assessing the strength of the linear relationship.

- A correlation of either −1 or +1 indicates that there is a perfect linear relationship, and all data points fall on the same straight line. If $r = +1$, there is perfect positive correlation. If $r = -1$, there is perfect negative correlation. In the real world, this will not generally happen.

- If $r = 0$ there is absolutely no linear relationship between *x* and *y*. (no linear correlation)
  A correlation of 0 indicates that the best straight line through the data is exactly horizontal, so knowing the value of *x* does not change the predicted value of *y*.

**What the SIGN of "r" (sample correlation coefficient) tells us:**

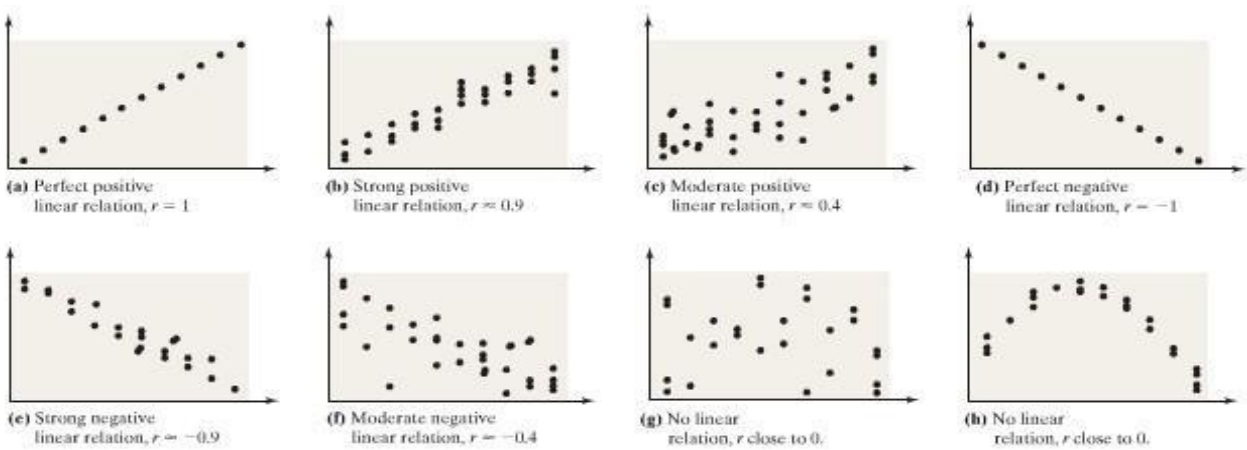
- The *sign* of the correlation indicates the direction of the relationship.

- A *positive* correlation indicates that the two variables tend to increase together. (As x increases, y tends to increase).

- A *negative* correlation indicates that when one variable increases, the other is likely to decrease. (As x increases, y tends to decrease).

- The sign or "r" is the same as the sign of the slope, $b_1$

| |
|---|
| Strong correlation **does not** suggest that x causes y or y causes x. There is simply a "potential" of increase or decrease.<br>**CORRELATION DOES NOT IMPLY CAUSATION!** |




**Common Errors Involving Correlation:**

- Correlation is very sensitive to outliers, so look at the scatterplot.
- Positive correlations imply positive slopes, but **large correlations do not imply large slopes.**
- The lack of a linear relationship does not imply a lack of a relationship in general.
- CORRELATION DOES NOT IMPLY CAUSATION!

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**Rule of thumb for interpreting values of *r*:**

If $|r| < 0.25$, conclude no linear correlation.

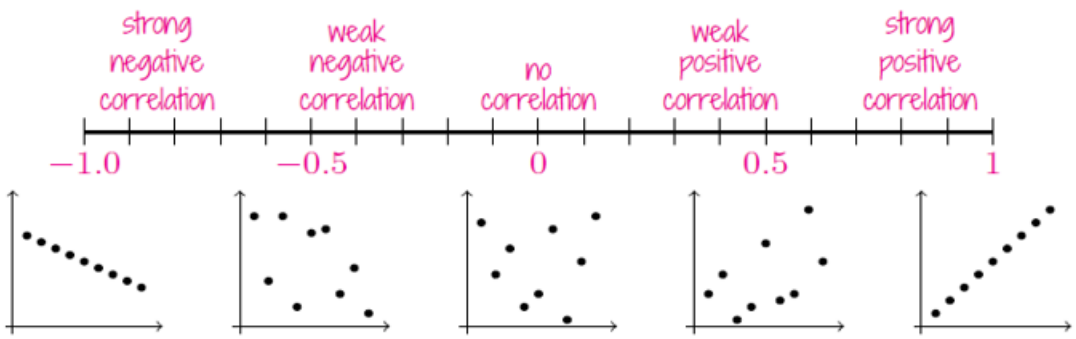
$0.25 < |r| < 0.50$, conclude weak (positive/negative) linear correlation.

$0.50 < |r| < 0.75$, conclude moderate (positive/negative) linear correlation.

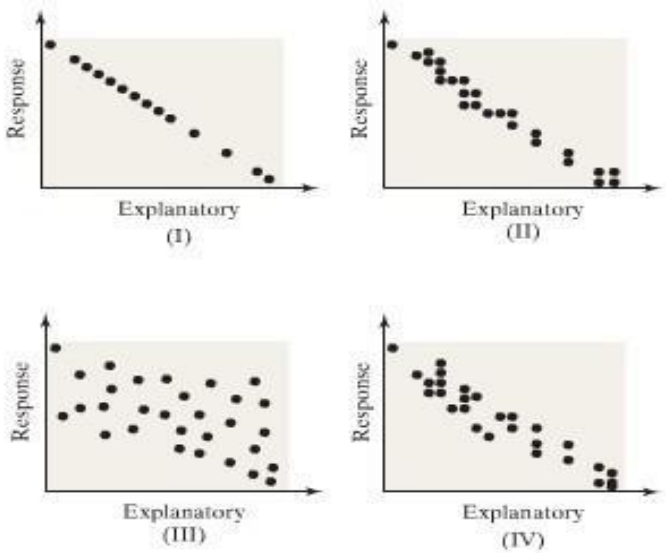
$|r| > 0.75$, conclude strong (positive/negative) linear correlation.

**The Correlation Coefficient**

The _____, denoted by ___, is a quantity that measures the _____ and the _____ of the linear association between two variables. "r" is in the interval _____.




Example 3. Match the linear correlation coefficient to the scatter diagram. The scales on the *x*- and *y*-axes are the same for each scatter diagram.




*(a)* r = −0.969

*(b)* r = −0.049

*(c)* r = −1

*(d)* r = −0.992

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## 12.3: The Regression Equation

Data rarely fits a straight line exactly. Usually, you must be satisfied with rough predictions. Typically, you have a set of data whose scatter plot appears to "**fit**" a straight line. This is called a **Line of Best Fit** or **Least-Squares Line**.

**Regression analysis** is the area of statistics that is used to examine the relationship between a quantitative response variable and one or more explanatory variables.

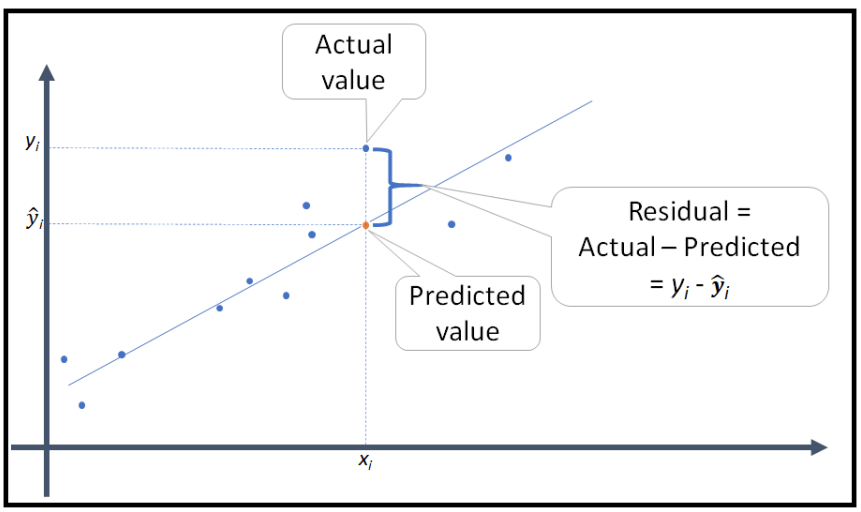
**Linear regression equation (best fit line)** - an algebraic description of the linear relationship between two variables, defined by

$$\hat{y} = b_1 x + b_0.$$

Where $x$ is the explanatory variable, $\hat{y}$ is the PREDICTED response variable, $b_0$ is the $y$-intercept, and $b_1$ is the slope.

- The equation can be used in the future to **predict** values of a response variable when we only know the values for the explanatory variable.
- All the points will likely not fall on the line in a regression equation, which means, the actual $y$ will often differ from the predicted $y$ ($\hat{y}$).

**Residual** - the difference between the actual observed sample value, $y$, and the predicted value of $y$, $\hat{y}$, from the regression equation, computed as $y - \hat{y}$ (observed $_y$ – predicted $_y$).




**Least-squares regression line** - the line that minimizes the sum of the squared errors (or residuals), or rather the sum of the squared vertical distance between the observed values of $y$ and those predicted by the line, $\hat{y}$.

**Guidelines for Using the Regression Equation for Predictions:**

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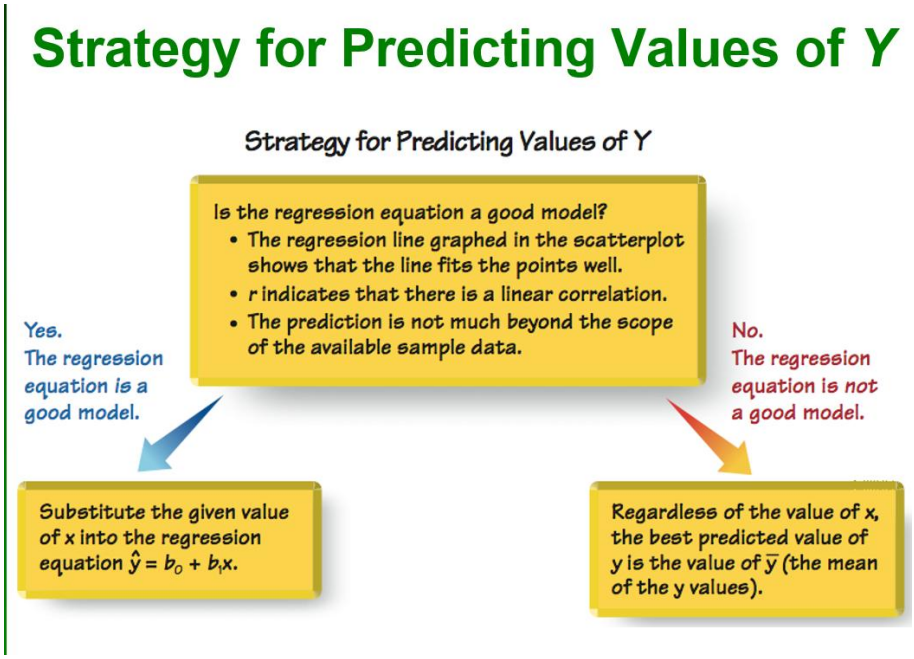
- Use the regression equation for predictions only if the graph of the regression line fits the points reasonably well.

- Use the regression equation for predictions only if *r* indicates there is a linear correlation present.

- Use the regression equation for predictions only if the data do not go much beyond the scope of the available sample data.




**Excel Directions for Determining the Least-Squares Regression Line:**

1. Enter the explanatory variable in column A.

2. Enter the response variable in column B.

3. With the cursor in the Input Y-Range, select the column that contains the response variable (column B).

4. With the cursor in the Input X-Range, select the column that contains the explanatory variable (column A).

5. Select Output Range, and click on an empty cell, then select OK.

Example 4. The Excel Output from Example 1 (golfing)

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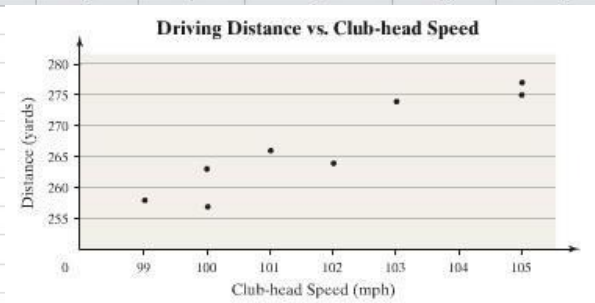
SUMMARY OUTPUT

| Regression Statistics | |
|---|---|
| Multiple R | 0.938695838 |
| R Square | 0.881149876 |
| Adjusted R Square | 0.861341522 |
| Standard Error | 2.882638465 |
| Observations | 8 |




ANOVA

| | df | SS | MS | F | Significance F |
|---|---|---|---|---|---|
| Regression | 1 | 369.6423729 | 369.64237 | 44.4837503 | 0.000549825 |
| Residual | 6 | 49.85762712 | 8.3096045 | | |
| Total | 7 | 419.5 | | | |

| | Coefficients | Standard Error | t Stat | P-value | Lower 95% | Upper 95% | Lower 95.0% | Upper 95.0% |
|---|---|---|---|---|---|---|---|---|
| Intercept | -55.7966102 | 48.37134953 | -1.153505 | 0.29257431 | -174.1570386 | 62.563818 | -174.1570386 | 62.56381826 |
| X Variable 1 | 3.166101695 | 0.47470539 | 6.669614 | 0.00054983 | 2.00453945 | 4.3276639 | 2.00453945 | 4.32766394 |

What is the correlation coefficient?

What is the slope of the regression line?

What is the y-intercept of the regression line?

What is the equation of the regression line (the prediction line)?

## Prediction Errors and Residuals

Before the use of Excel, a regression line was created by hand. The figure below shows this line, and the regression line was found to be

$$\hat{y} = -22.4967 + 2.833x \text{ (found by hand – not Excel)}$$

We found a line that fits the data, but is this the ***best*** line?

We need a criterion for determining "**best.**" Consider the figure on the next page.

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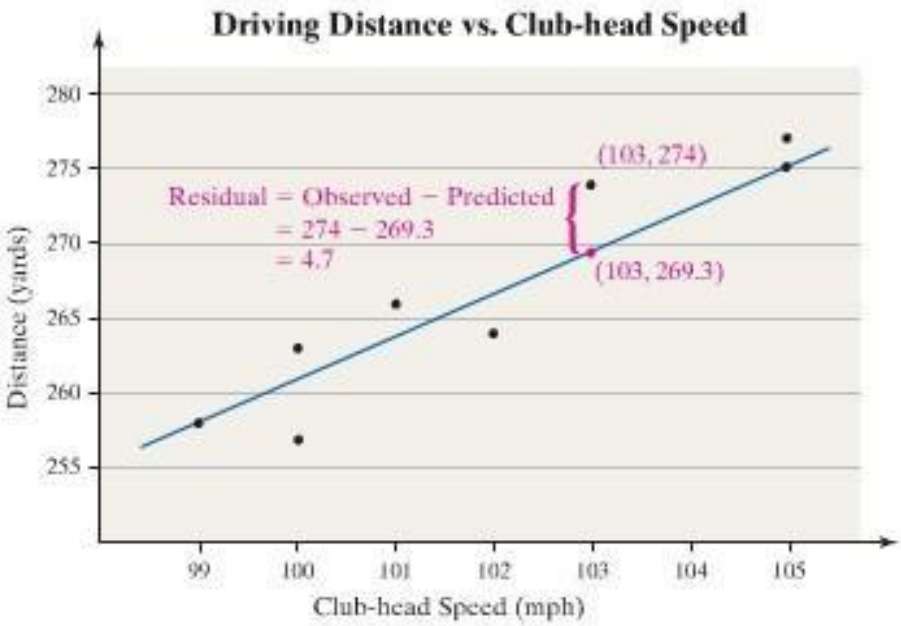


Each *y*-coordinate on the line corresponds to a predicted distance for a given club-head speed. For example, if club-head speed is 103 miles per hour, the predicted distance the ball will travel is 2.833(103)−22.4967 = 269.3 yards. The observed distance for this club-head speed is 274 yards. The difference between the observed value of *y* and the predicted value of *y* is the error, or **residual**. For a club-head speed of 103 miles per hour, the residual is

$$\begin{aligned}\text{Residual} &= \text{observed } y - \text{predicted } y\\ &= y - \hat{y}\\ &= 274 - 269.3\\ &= 4.7 \text{ yards}\end{aligned}$$

The **residual** or **prediction error**, for a club-head speed of 103 miles per hour is 4.7 yards. This is shown in the figure above.

## INTERPRET THE SLOPE AND *y*-INTERCEPT OF THE LEAST-SQUARES REGRESSION LINE

The definition of the slope of a line is $\frac{Rise}{Run}$ or $\frac{Change\ in\ y}{Change\ in\ x}$. For a line whose slope is $\frac{2}{3}$, if *x* increases by 3, *y* will *increase* by 2. Of course, if the slope of a line is $-4 = \frac{-4}{1}$ if *x* increases by 1, *y* will *decrease* by 4.

The *y*-intercept of any line is the point where the graph intersects the vertical axis. It is found by letting *x* = 0 in an equation and solving for *y*.

From Example 4 Excel output, we found the regression equation to be

$$\hat{y} = 3.1661x - 55.7966.$$

**Interpretation of Slope:**

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The slope of the regression line is 3.1661. If the club-head speed increases by 1 mile per hour, the distance the ball travels will increase by 3.1661 yards, **on average**. We use the phrase "on average" because we cannot say an increase in swing speed of 1 mile per hour guarantees the distance will increase 3.1661 yards. The slope represents what we expect the response variable (distance) to change by for a 1-unit change in the explanatory variable (swing speed).

| |
|---|
| For a one unit increase in [the explanatory variable], [the response variable] will [increase/decrease] by [the value of the slope], on average. |

**Interpretation of the y-intercept:**

The *y*-intercept of the regression line is −55.7966. To interpret the *y*-intercept, we must first ask two questions:

1. Is 0 a reasonable value for the explanatory variable? (Can *x*=0 in the problem?)
2. Do any observations near *x* = 0 exists in the data set?

If the answer to either of these questions is no, we do not give an interpretation to the *y*-intercept. In the regression equation in Example 4, a swing speed of 0 miles per hour does not make sense, so an interpretation of the *y*-intercept is unreasonable. (no practical interpretation)

| |
|---|
| In general, to interpret a *y*-intercept, we would say that it is the expected value of the response variable when the value of the explanatory variable is 0. Remember though, for many problems, there will be NO PRACTICAL INTERPRETATION of the y-intercept. |

**Interpreting the Coefficient of Determination, $r^2$:**

The variable $r^2$ is called the **coefficient of determination** and is the square of the correlation coefficient. It is usually stated as a percent, rather than decimal form.

$r^2$, when expressed as a percent, represents the percent of variation in the dependent (predicted) variable *y* that can be explained by the variation in the independent (explanatory) variable *x* using the regression (best fit) line.

| |
|---|
| The **coefficient of determination,** $r^2$, measures the proportion of total variation in the response variable that is explained by the least-squares regression line. |

**Notes about the Coefficient of Determination**

$R^2$ is always between 0 and 1. That is, $0 \le r^2 \le 1$.

A coefficient of determination of 1 indicates that the least squares regression line explains 100% of the variation in the response variable (*y*).

A coefficient of determination of 0 indicates that the least squares regression line explains none of the variation in the response variable (*y*).

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Determining the Coefficient of Determination Using Technology

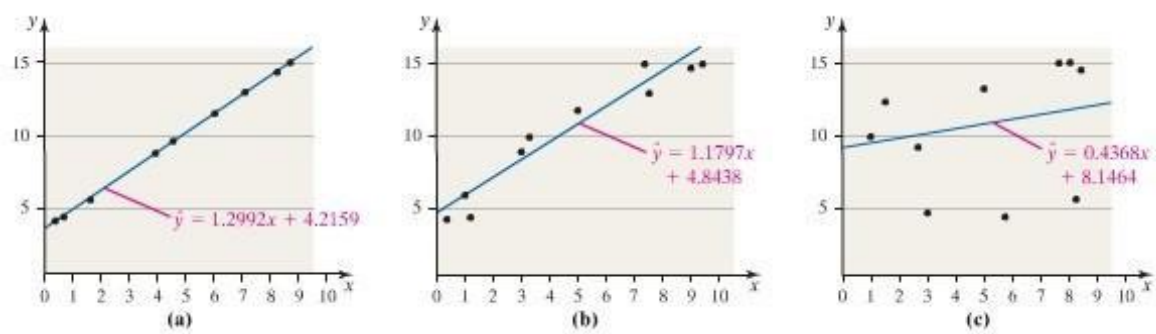
The coefficient of determination, $r^2$, is found by using the same Excel steps given at the bottom of page 8.

| SUMMARY OUTPUT | |
|---|---|
| | |
| *Regression Statistics* | |
| Multiple R | 0.938695838 |
| R Square | 0.881149876 |
| Adjusted R Square | 0.861341522 |
| Standard Error | 2.882638465 |
| Observations | 8 |

Consider the three data sets given below.

| Data Set A | | Data Set B | | Data Set C | |
|---|---|---|---|---|---|
| *x* | *y* | *x* | *y* | *x* | *y* |
| 3.6 | 8.9 | 3.1 | 8.9 | 2.8 | 8.9 |
| 8.3 | 15.0 | 9.4 | 15.0 | 8.1 | 15.0 |
| 0.5 | 4.8 | 1.2 | 4.8 | 3.0 | 4.8 |
| 1.4 | 6.0 | 1.0 | 6.0 | 8.3 | 6.0 |
| 8.2 | 14.9 | 9.0 | 14.9 | 8.2 | 14.9 |
| 5.9 | 11.9 | 5.0 | 11.9 | 1.4 | 11.9 |
| 4.3 | 9.8 | 3.4 | 9.8 | 1.0 | 9.8 |
| 8.3 | 15.0 | 7.4 | 15.0 | 7.9 | 15.0 |
| 0.3 | 4.7 | 0.1 | 4.7 | 5.9 | 4.7 |
| 6.8 | 13.0 | 7.5 | 13.0 | 5.0 | 13.0 |

The figure below represents the scatter diagrams for each data set.




| Data | $r^2$ | Interpretation |
|---|---|---|
| A | 99.99% | 99.99% of the variability in *y* is explained by the least-squares regression line. |
| B | 94.7% | 94.7% of the variability in *y* is explained by the least-squares regression line. |
| C | 9.4% | 9.4% of the variability in *y* is explained by the least-squares regression line. |

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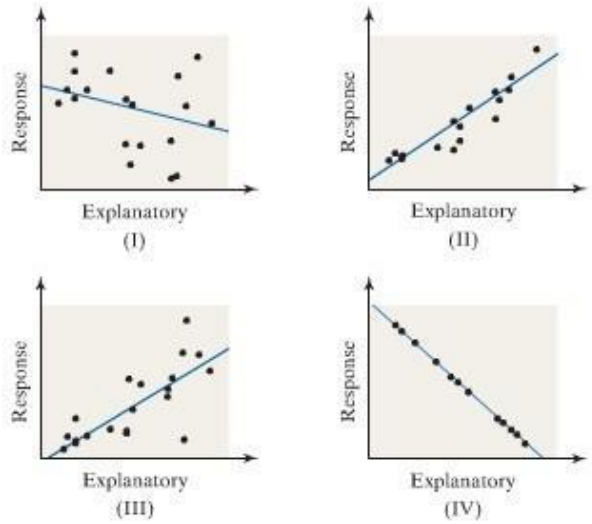
Example 5. Match the coefficient of determination to the scatter diagram. The scales on the horizontal and vertical axis are the same for each scatter diagram.

(a) $r^2 = 0.58$
(b) $r^2 = 0.90$
(c) $r^2 = 1$
(d) $r^2 = 0.12$




Example 6. A student at Joliet Junior College conducted a survey of 20 randomly selected full-time students to determine the relation between the number of hours of video game playing each week, $x$, and grade-point average, $y$. She found that a linear relation exists between the two variables. The least-squares regression line that describes this relation is
$\hat{y} = -0.0526x + 2.9342$. Note that $R^2$ is equal to 0.372.

(a) Find and interpret the linear correlation coefficient.

(b) Predict the grade-point average of a student who plays video games 8 hours per week.

(c) Interpret the slope.

(d) If appropriate, interpret the $y$-intercept.

(e) A student who plays video games 7 hours per week has a grade-point average of 2.68. What is the residual for a student who plays video games 7 hours per week?

Example 7. You and your friends go to your favorite frozen yogurt place and the price is determined by the weight of the container. The frozen yogurt costs 50 cents per ounce. You and your friends fill

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a container that weighs 5 ounces when it is empty. Define *x* to be the weight of the filled container (in ounces) and y to be the price you or your friends are charged (in dollars). The relationship is $\hat{y} = -2.50 + 0.50x,$ and the linear correlation is given as 1.

(a) Find and interpret *r*.

(b) Calculate $R^2$ and write a sentence that interprets this value in the context of this situation.

(c) Explain (in words) what it would mean if *x* = 5.

(d) If appropriate, interpret the *y*-intercept.

(e) Interpret the slope.

(f) If the container plus ingredients weighs 20 ounces, how much does the yogurt cost your friend?

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Example 8. The American black bear is one of eight bear species in the world. It is the smallest North American bear and the most common bear species on the planet. In 1969, Dr. Michael R. Pelton of the University of Tennessee initiated a long-term study of the population in the Great Smoky Mountains National Park. One aspect of the study was to develop a model that could be used to predict a bear's weight (since it is not practical to weigh bears in the field). One variable thought to be related to weight is the length of the bear. The following data represent the lengths and weights of 12 American black bears. Use the excel output on the bottom of the page.

| Total Length (cm) | 139.0 | 138.0 | 139.0 | 120.5 | 149.0 | 141.0 | 141.0 | 150.0 | 166.0 | 151.5 | 129.5 | 150.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Weight (kg) | 110 | 60 | 90 | 60 | 85 | 110 | 95 | 85 | 155 | 140 | 105 | 110 |

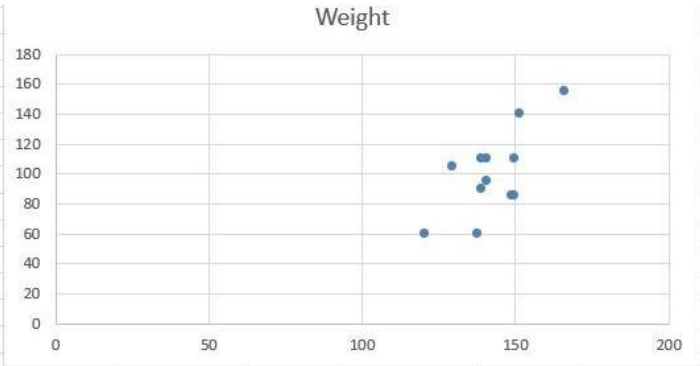
(a) Which variable is the explanatory variable based on the goals of the research?

(b) Determine the regression line between weight and height.

(c) Predict the weight of a bear if the height is 155.5 cm.

(d) Find and interpret *r*.

(e) Compute and interpret the coefficient of determination.




SUMMARY OUTPUT

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.694653357 |
| R Square | 0.482543287 |
| Adjusted R Square | 0.430797615 |
| Standard Error | 21.24554492 |
| Observations | 12 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 1 | 4209.184878 | 4209.184878 | 9.325288 | 0.012173922 |
| Residual | 10 | 4513.731788 | 451.3731788 | | |
| Total | 11 | 8722.916667 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* | *Lower 95.0%* | *Upper 95.0%* |
|---|---|---|---|---|---|---|---|---|
| Intercept | -139.8379309 | 78.91437901 | -1.772020925 | 0.1068017 | -315.6701247 | 35.994263 | -315.6701247 | 35.99426296 |
| Length | 1.681571986 | 0.550661023 | 3.053733449 | 0.0121739 | 0.454622766 | 2.90852121 | 0.454622766 | 2.908521207 |

Example 9. A researcher collects data on imported cars considered to be lemons (a car with defects that make it unsafe to drive) and the crash fatality rate in the USA. The seven observations are randomly selected cities across the country.

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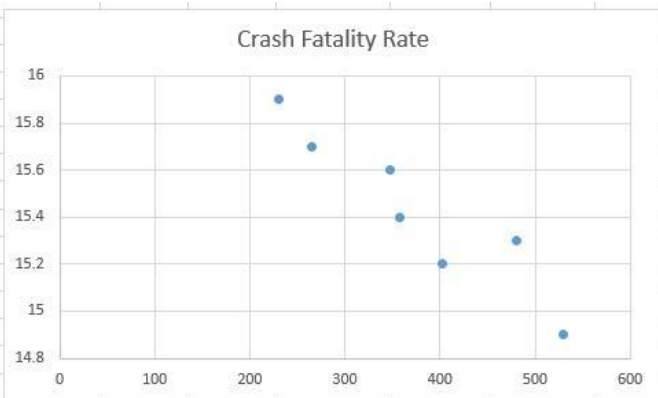


SUMMARY OUTPUT

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.939894457 |
| R Square | 0.88340159 |
| Adjusted R Square | 0.860081907 |
| Standard Error | 0.125395887 |
| Observations | 7 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 1 | 0.595665072 | 0.595665 | 37.88223 | 0.001646686 |
| Residual | 5 | 0.078620642 | 0.015724 | | |
| Total | 6 | 0.674285714 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* | *Lower 95.0%* | *Upper 95.0%* |
|---|---|---|---|---|---|---|---|---|
| Intercept | 16.51743501 | 0.18315002 | 90.18528 | 3.18E-09 | 16.04663289 | 16.988237 | 16.0466329 | 16.9882371 |
| Lemon Imports | -0.002918088 | 0.000474112 | -6.15485 | 0.001647 | -0.00413683 | -0.0016993 | -0.0041368 | -0.00169935 |

What is the regression equation?

A. $\hat{y} = 16.517 - 0.003x$
B. $\hat{y} = 16.517x - 0.003$
C. $y = 16.517 - 0.003x$
D. $y = 16.517x - 0.003$

The best description of the correlation is to describe it as

A. strong positive correlation.
B. strong linear correlation.
C. strong negative correlation.
D. None of the above

Interpret the slope of the line.

A. For a 1 unit increase in the crash fatality rate, lemon imports will, on average, increase by 16.517 units.
E. For a 1 unit increase in lemon imports, the crash fatality rate will, on average, increase by 0.003 units.
F. For a 1 unit increase in lemon imports, the crash fatality rate will, on average, increase by 16.517 units.
G. For a 1 unit increase in the crash fatality rate, lemon imports will, on average, increase by 0.003 units.
H. None of the above

Give a practical interpretation of the coefficient of determination.

A. We can predict the crash fatality rate correctly 88.34% of the time using lemon imports in a least-squares regression line.
B. 93.99% of the differences in crash fatality rates are caused by differences in lemon imports.

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C. 93.99% of the sample variation in crash fatality rates can be explained by the least-squares regression line.
D. We can predict the crash fatality rate correctly 93.99% of the time using lemon imports in a least-squares regression line.
E. 88.34% of the differences in crash fatality rates are caused by differences in lemon imports.
F. 88.34% of the sample variation in crash fatality rates can be explained by the least-squares regression line.

Is it reasonable to use the regression equation to make a prediction for lemon imports of 100? Justify your answer.

A. Yes, all of the criteria are met.
B. No, $r$ does not indicate that there is a reasonable amount of correlation.
C. No, this prediction is far outside the scope of available data.
D. No, the regression line does not fit the points reasonably well.

What can we say about the relationship between lemon imports and crash fatality rate?

A. The is no relationship between lemon imports and crash fatality rate.
B. While lemon imports and crash fatality rate are strongly related, this relationship is likely due to some third variable that affects them both.
C. Because of the strong relationship, lemon imports are a leading cause of crash fatality rate.

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